CSWP: march on in time

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December 2025

1 Summary of the problem

In this assignment we implement the march-on in time algorithm on PEC scatterers. We implemented on TM case of the wave equation. This means that the electric field only contains a z-component and the magnetic field only in the x and y- direction. the problem is also consistent in the z-direction this means the material is symmetric under the z-axis. This reduces our problem to a 2 dimensional problem.

After that we use the greens function in its retarded form to solve the boundary integral equation which has as the unknown the current. At last we discretize the integral equation as done in the assignment. And we us those formulae to describe the different problems.

2 assignment

we validate our code by calculating the time domain scattering at a cylinder.

The formula description is done in the assignment. Our task is implement the march-on-in-time algorithm on this example and look if this correspondds with the analytical solution.

for the first part we calculate the timedomain scattering at a cylinder. Using an incoming wave going to the positive x direction.

3 Implementation

The algorithm is implemented in Python, using the Numpy and Scipy packages.

The MOT algorithm is fully contained in the `MOT` class, initiating an instance takes a few arguments:

- `curve` is a `numpy` array of shape $(2, N_S + 1)$. The first index is the coordinate (`0` for x or `1` for y, generally `_x` and `_y` were written as aliases for 0 and 1). N_S is the number of segments in which the curve has been

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Table 1: Parameters for validation

discretized. This field consists out of the edges of these segments (the first and last elements must equal),

- `dt` is the time step in seconds,
- `N_T` is the number of timesteps taken, and
- `N_G` the number of points for the Gaussian quadrature.

With these arguments the quadrature points (`coordquad` and `weights`), the geometry fields (`rho` are the midpoints, `tangents` the tangent vectors and `l` the lengths of the segments, `rhop[_x, m, g]` is the x-coordinate of the m-th segment, linearly interpolated for x_g), F (an array of shape (N_T, N_S, N_S, N_G), such that $F(k, \rho_m, \rho_{n,g}) =$ `F[k,m,n,g]`) and Z (with shape (N_T, N_S, N_S), so that $(Z_k)_{m,n} =$ `Z[k,m,n]`) are calculated.

Finally the function `solve` implements the stepping algorithm. It takes one argument `V`, which is an array of shape (N_T, N_S), such that $(V_j)_m = E^i(\rho_m, j\Delta t) =$ `V[j,m]`. The solver (and `_init_Z`) uses the Numpy function `einsum`, which makes translating sums from math into code a lot easier:

$$\sum_{g}^{N_G} w_g F(k, \rho_m, \rho_{n,g}) = \texttt{np.einsum('g,kmng->kmn', weights, F)}$$

This and a few reshapes which had to happen for Numpy to be able to correctly broadcast together some arrays, are the only noteworthy difference between the code and its mathematic description from the assignment.

TODO maybe some stuff about the second excercise and the E^i calculation.

4 Validation

The spectrum of the incoming wave quickly decays to zero. This means that when dividing by this spectrum the result will explode. Due to this the frequency range for which the solver can be used is only limited. We chose to limit the frequency range to the region where the spectrum of the incoming wave is more than 0.1 % of its maximum.

When calculating the Fourier transform of the solution from the MOT-algorithm, Numpy's `rfft` function was used. It is important to note that this is a discrete transform without any normalization factor. To compare it with the continuous transform, a correction factor of $dt/\sqrt{2\pi}$ is needed.

TODO plot of (incident?) electric field and induced current (note/show that the electric field is indeed zero at time 0 at the PEC)

4.1 Analytical Solution

TODO reproduction of fig 7 (basically done, just missing correct analytical current) + discuss influence of simulation params on freq range

5 Resonance Freqeuencies of Cylindrical Cavity

5.1 Incident Field by Numerical Quadrature

5.2 Resonance Modes

5.3 Spectrum Numerical Solution

5.4 Off-center Source

6 Creative Assignment